

maxima) in the spectrum obtained with the substance to be examined correspond in position and relative size to those in the spectrum obtained with the reference substance (CRS).

When the spectra recorded in the solid state show differences in the positions of the transmission minima (absorption maxima), treat the substance to be examined and the reference substance in the same manner so that they crystallise or are produced in the same form, or proceed as prescribed in the monograph, then record the spectra.

IDENTIFICATION USING REFERENCE SPECTRA

Control of resolution performance. For instruments having a monochromator, record the spectrum of a polystyrene film approximately 35 µm in thickness. The difference x (see Figure 2.2.24.-1) between the percentage transmittance at the transmission maximum A at 2870 cm⁻¹ (3.48 µm) and that at the transmission minimum B at 2849.5 cm⁻¹ (3.51 µm) must be greater than 18. The difference y between the percentage transmittance at the transmission maximum C at 1589 cm⁻¹ (6.29 µm) and that at the transmission minimum D at 1583 cm⁻¹ (6.32 µm) must be greater than 10.

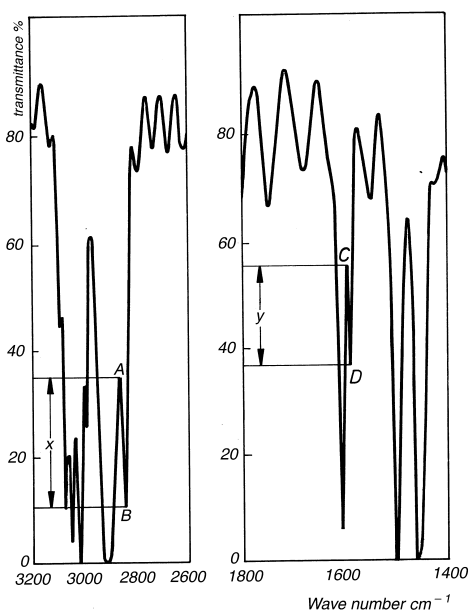


Figure 2.2.24.-1. – Typical spectrum of polystyrene used to verify the resolution performance

For Fourier-transform instruments, use suitable instrument resolution with the appropriate apodisation prescribed by the manufacturer. The resolution is checked by suitable means, for example by recording the spectrum of a polystyrene film approximately 35 µm in thickness. The difference between the absorbances at the absorption minimum at 2870 cm⁻¹ and the absorption maximum at 2849.5 cm⁻¹ is greater than 0.33. The difference between the absorbances at the absorption minimum at 1589 cm⁻¹ and the absorption maximum at 1583 cm⁻¹ is greater than 0.08.

Verification of the wave-number scale. The wave-number scale may be verified using a polystyrene film, which has transmission minima (absorption maxima) at the wave numbers (in cm⁻¹) shown in Table 2.2.24.-1.

Table 2.2.24.-1. – Transmission minima and acceptable tolerances of a polystyrene film

Transmission minima (cm ⁻¹)	Acceptable tolerance (cm ⁻¹)	
	Monochromator instruments	Fourier-transform instruments
3060.0	± 1.5	± 1.0
2849.5	± 2.0	± 1.0
1942.9	± 1.5	± 1.0
1601.2	± 1.0	± 1.0
1583.0	± 1.0	± 1.0
1154.5	± 1.0	± 1.0
1028.3	± 1.0	± 1.0

Method. Prepare the substance to be examined according to the instructions accompanying the reference spectrum/reference substance. Using the operating conditions that were used to obtain the reference spectrum, which will usually be the same as those for verifying the resolution performance, record the spectrum of the substance to be examined.

The positions and the relative sizes of the bands in the spectrum of the substance to be examined and the reference spectrum are concordant in the 2 spectra.

Compensation for water vapour and atmospheric carbon dioxide. For Fourier-transform instruments, spectral interference from water vapour and carbon dioxide is compensated using suitable algorithms according to the manufacturer's instructions. Alternatively, spectra can be acquired using suitable purged instruments or ensuring that sample and background single beam spectra are acquired under exactly the same conditions.

IMPURITIES IN GASES

For the analysis of impurities, use a cell transparent to infrared radiation and of suitable optical path length (for example, 1-20 m). Fill the cell as prescribed under Gases. For detection and quantification of the impurities, proceed as prescribed in the monograph.

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2.2.25. ABSORPTION SPECTROPHOTOMETRY, ULTRAVIOLET AND VISIBLE

Determination of absorbance. The absorbance (A) of a solution is defined as the logarithm to base 10 of the reciprocal of the transmittance (T) for monochromatic radiation:

$$A = \log_{10} \left(\frac{1}{T} \right) = \log_{10} \left(\frac{I_0}{I} \right)$$

$$T = I/I_0,$$

$$I_0 = \text{intensity of incident monochromatic radiation,}$$

$$I = \text{intensity of transmitted monochromatic radiation.}$$

In the absence of other physico-chemical factors, the measured absorbance (A) is proportional to the path length (b) through which the radiation passes and to the concentration (c) of the substance in solution in accordance with the equation:

$$A = \varepsilon \cdot c \cdot b$$

ε = molar absorptivity, if b is expressed in centimetres and c in moles per litre.

The expression $A_{1\text{ cm}}^{1\text{ per cent}}$ representing the specific absorbance of a dissolved substance refers to the absorbance of a 10 g/l solution in a 1 cm cell and measured at a defined wavelength so that:

$$A_{1\text{ cm}}^{1\text{ per cent}} = \frac{10\varepsilon}{M_r}$$

Unless otherwise prescribed, measure the absorbance at the prescribed wavelength using a path length of 1 cm and at $20 \pm 1^\circ\text{C}$. Unless otherwise prescribed, the measurements are carried out with reference to the same solvent or the same mixture of solvents. The absorbance of the solvent measured against air and at the prescribed wavelength shall not exceed 0.4 and is preferably less than 0.2. Plot the absorption spectrum with absorbance or function of absorbance as ordinate against wavelength or function of wavelength as abscissa.

Where a monograph gives a single value for the position of an absorption maximum, it is understood that the value obtained may differ by not more than ± 2 nm.

Apparatus. Spectrophotometers suitable for measuring in the ultraviolet and visible range of the spectrum consist of an optical system capable of producing monochromatic radiation in the range of 200 nm to 800 nm and a device suitable for measuring the absorbance.

Control of wavelengths. Verify the wavelength scale using the absorption maxima of *holmium perchlorate solution R*, the line of a hydrogen or deuterium discharge lamp or the lines of a mercury vapour shown in Table 2.2.25-1. The permitted tolerance is ± 1 nm for the ultraviolet range and ± 3 nm for the visible range.

Table 2.2.25-1. – Absorption maxima for control of wavelength scale

241.15 nm (Ho)	404.66 nm (Hg)
253.7 nm (Hg)	435.83 nm (Hg)
287.15 nm (Ho)	486.0 nm (D β)
302.25 nm (Hg)	486.1 nm (H β)
313.16 nm (Hg)	536.3 nm (Ho)
334.15 nm (Hg)	546.07 nm (Hg)
361.5 nm (Ho)	576.96 nm (Hg)
365.48 nm (Hg)	579.07 nm (Hg)

Control of absorbance. Check the absorbance using suitable filters or a solution of *potassium dichromate R* at the wavelengths indicated in Table 2.2.25-2 which gives for each wavelength the exact values and the permitted limits of the specific absorbance. The tolerance for the absorbance is ± 0.01 .

For the control of absorbance, use a solution of potassium dichromate prepared as follows. Dissolve 57.0 mg to 63.0 mg of *potassium dichromate R*, previously dried to constant mass at 130°C , in 0.005 M sulphuric acid and dilute to 1000.0 ml with the same acid.

Table 2.2.25-2

Wavelength (nm)	Specific absorbance $A_{1\text{ cm}}^{1\text{ per cent}}$	Maximum tolerance
235	124.5	122.9 to 126.2
257	144.5	142.8 to 146.2
313	48.6	47.0 to 50.3
350	107.3	105.6 to 109.0

Limit of stray light. Stray light may be detected at a given wavelength with suitable filters or solutions: for example the absorbance of a 12 g/l solution of *potassium chloride R* in a 1 cm cell increases steeply between 220 nm and 200 nm and is greater than 2 at a wavelength between 198 nm and 202 nm when compared with water as compensation liquid.

Resolution (for qualitative analysis). When prescribed in a monograph, measure the resolution of the apparatus as follows: record the spectrum of a 0.02 per cent V/V solution of *toluene R* in *hexane R*. The minimum ratio of the absorbance at the maximum at 269 nm to that at the minimum at 266 nm is stated in the monograph.

Spectral slit-width (for quantitative analysis). To avoid errors due to spectral slit-width, when using an instrument on which the slit-width is variable at the selected wavelength, the slit-width must be small compared with the half-width of the absorption band but it must be as large as possible to obtain a high value of I_0 . Therefore, a slit-width is chosen such that further reduction does not result in a change in absorbance reading.

Cells. The tolerance on the path length of the cells used is ± 0.005 cm. When filled with the same solvent, the cells intended to contain the solution to be examined and the compensation liquid must have the same transmittance. If this is not the case, an appropriate correction must be applied.

The cells must be cleaned and handled with care.

DERIVATIVE SPECTROPHOTOMETRY

Derivative spectrophotometry involves the transformation of absorption spectra (zero-order) into first-, second- or higher-order-derivative spectra.

A *first-derivative spectrum* is a plot of the gradient of the absorption curve (rate of change of the absorbance with wavelength, $dA/d\lambda$) against wavelength.

A *second-derivative spectrum* is a plot of the curvature of the absorption spectrum against wavelength ($d^2A/d\lambda^2$). The second derivative at any wavelength λ is related to concentration by the following equations:

$$\frac{d^2A}{d\lambda^2} = \frac{d^2A_{1\text{ cm}}^{1\text{ per cent}}}{d\lambda^2} \times \frac{c'b}{10} = \frac{d^2A\varepsilon}{d\lambda^2} \times \frac{cb}{10}$$

c' = concentration of the absorbing solute, in grams per litre.

Apparatus. Use a spectrophotometer complying with the requirements prescribed above and equipped with an analogue resistance-capacitance differentiation module or a digital differentiator or other means of producing derivative spectra. Some methods of producing second-derivative spectra produce a wavelength shift relative to the zero-order spectrum and this is to be taken into account where applicable.

Resolution power. When prescribed in a monograph, record the second-derivative spectrum of a 0.2 g/l solution of *toluene R* in *methanol R*, using *methanol R* as the compensation liquid. The spectrum shows a small negative extremum located between 2 large negative extrema at 261 nm and 268 nm, respectively, as shown in Figure 2.2.25.-1. Unless otherwise prescribed in the monograph, the ratio A/B (see Figure 2.2.25.-1) is not less than 0.2.

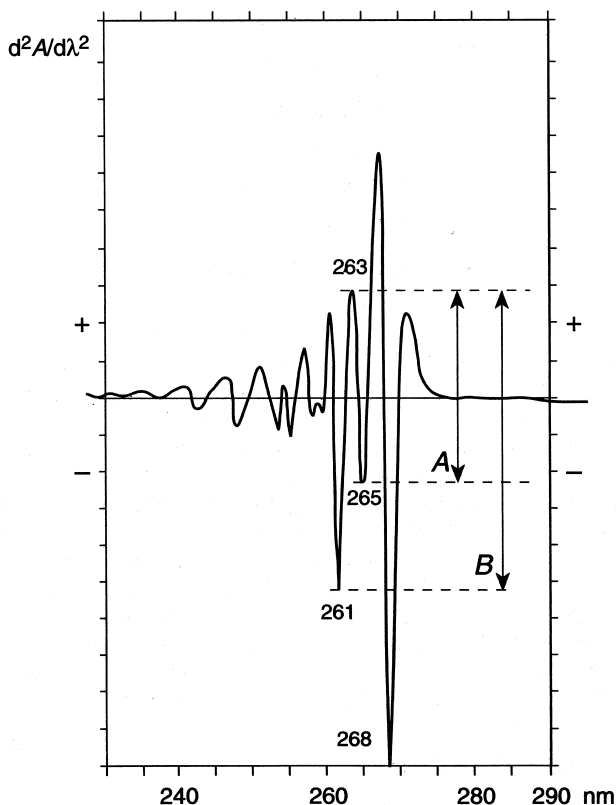


Figure 2.2.25.-1

Procedure. Prepare the solution of the substance to be examined, adjust the various instrument settings according to the manufacturer's instructions and calculate the amount of the substance to be determined as prescribed in the monograph.

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2.2.26. PAPER CHROMATOGRAPHY

ASCENDING PAPER CHROMATOGRAPHY

Apparatus. The apparatus consists of a glass tank of suitable size for the chromatographic paper used, ground at the top to take a closely fitting lid. In the top of the tank is a device which suspends the chromatographic paper and is capable of being lowered without opening the chamber. In the bottom of the tank is a dish to contain the mobile phase into which the paper may be lowered. The chromatographic paper consists of suitable filter paper, cut into strips of sufficient length and not less than 2.5 cm wide; the paper is cut so that the mobile phase runs in the direction of the grain of the paper.

Method. Place in the dish a layer 2.5 cm deep of the mobile phase prescribed in the monograph. If prescribed in the monograph, pour the stationary phase between the walls of the tank and the dish. Close the tank and allow to stand for 24 h at 20 °C to 25 °C. Maintain the tank at this temperature throughout the subsequent procedure. Draw a fine pencil

line horizontally across the paper 3 cm from one end.

Using a micro pipette, apply to a spot on the pencil line the volume of the solution prescribed in the monograph. If the total volume to be applied would produce a spot more than 10 mm in diameter, apply the solution in portions allowing each to dry before the next application. When more than one chromatogram is to be run on the same strip of paper, space the solutions along the pencil line at points not less than 3 cm apart. Insert the paper into the tank, close the lid and allow to stand for 1 h 30 min. Lower the paper into the mobile phase and allow elution to proceed for the prescribed distance or time. Remove the paper from the tank and allow to dry in air. Protect the paper from bright light during the elution process.

DESCENDING PAPER CHROMATOGRAPHY

Apparatus. The apparatus consists of a glass tank of suitable size for the chromatographic paper used, ground at the top to take a closely fitting glass lid. The lid has a central hole about 1.5 cm in diameter closed by a heavy glass plate or a stopper. In the upper part of the tank is suspended a solvent trough with a device for holding the chromatographic paper. On each side of the trough, parallel to and slightly above its upper edges, are two glass guide rods to support the paper in such a manner that no part of it is in contact with the walls of the tank. The chromatographic paper consists of suitable filter paper, cut into strips of sufficient length, and of any convenient width between 2.5 cm and the length of the trough; the paper is cut so that the mobile phase runs in the direction of the grain of the paper.

Method. Place in the bottom of the tank a layer 2.5 cm deep of the solvent prescribed in the monograph, close the tank and allow to stand for 24 h at 20 °C to 25 °C. Maintain the tank at this temperature throughout the subsequent procedure. Draw a fine pencil line horizontally across the paper at such a distance from one end that when this end is secured in the solvent trough and the remainder of the paper is hanging freely over the guide rod, the line is a few centimetres below the guide rod and parallel with it. Using a micro-pipette, apply on the pencil line the volume of the solution prescribed in the monograph. If the total volume to be applied would produce a spot more than 10 mm in diameter, apply the solution in portions, allowing each to dry before the next application. When more than one chromatogram is to be run on the same strip of paper, space the solutions along the pencil line at points not less than 3 cm apart. Insert the paper in the tank, close the lid, and allow to stand for 1 h 30 min. Introduce into the solvent trough, through the hole in the lid, a sufficient quantity of the mobile phase, close the tank and allow elution to proceed for the prescribed distance or time. Remove the paper from the tank and allow to dry in air. The paper should be protected from bright light during the elution process.

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2.2.27. THIN-LAYER CHROMATOGRAPHY

Thin-layer chromatography is a separation technique in which a stationary phase consisting of an appropriate material is spread in a uniform thin layer on a support (plate) of glass, metal or plastic. Solutions of analytes are deposited on the plate prior to development. The separation is based on adsorption, partition, ion-exchange or on combinations of these mechanisms and is carried out by migration (development) of solutes (solutions of analytes) in a solvent or a suitable mixture of solvents (mobile phase) through the thin-layer (stationary phase).